

Seals in Motion.

Flexible Sealing Solutions for the Mobility of the Future.



Introduction to AI-Driven Quality Control
Inspection in Wallstabe & Schneider

Niederwinkling, 18.04.23

Wen-Hao Wu

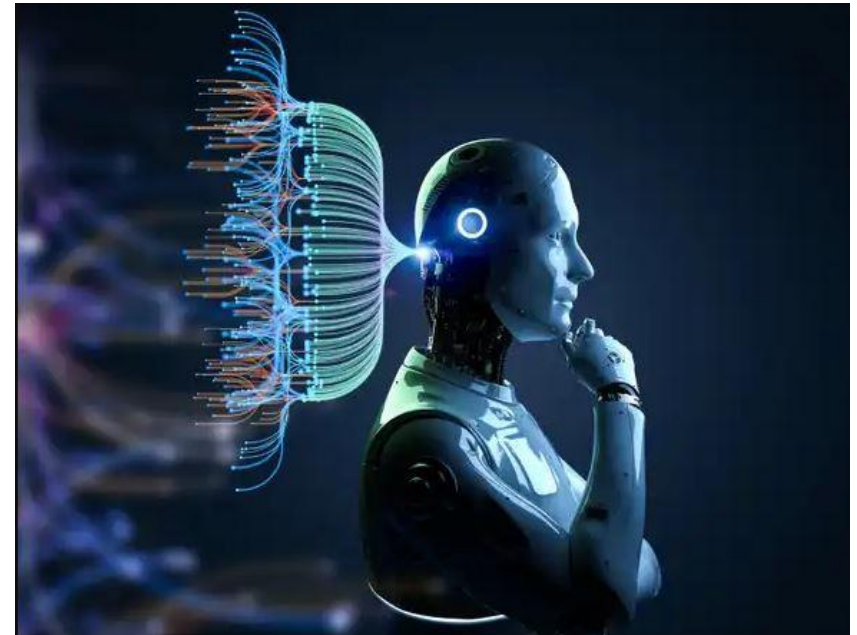
Agenda

- Fundamentals of AI
 - What is AI?
 - Why do we need AI?
 - What kinds of AI are there?
 - How does AI learn?
- Demo

Fundamentals of AI

What is Artificial Intelligence?

- Machines or systems that can **perform tasks and make decisions that typically require human intelligence**



Source: <https://seekingalpha.com/article/4593012-the-softer-side-of-artificial-intelligence>

Fundamentals of AI

Why do we need AI?

- Detect defects to **ensure product quality**
- **automate** the inspection process and **improve accuracy**



Fundamentals of AI

What kinds of AI are there?

- Natural Language Processing (Alexa, Siri)
- Robotics (Cleaning Robot)
- **Computer Vision** (Face-recognition, defect pattern recognition)
- ...



Natural Language Processing



Robotics

Fundamentals of AI

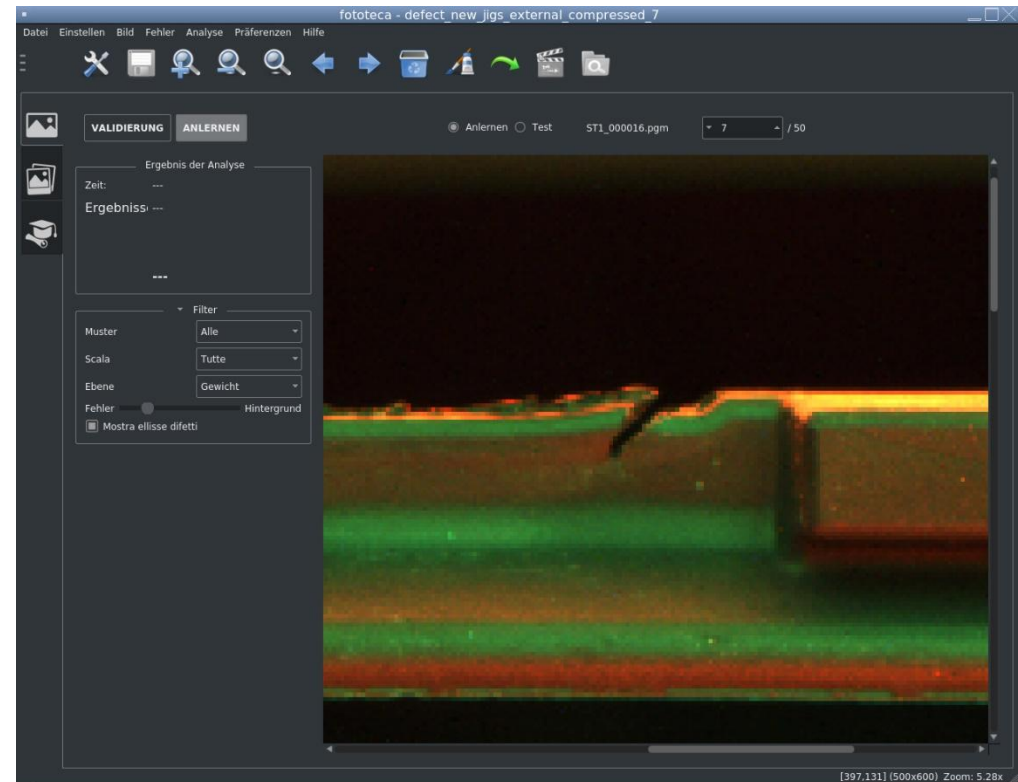
How does AI learn?

- AI learns from (high quality) **data**
- The „data“ it learns from, in our case, are **product pictures**.
- The method we used is so called „**Supervised Learning**“
- Training with **labeled data**



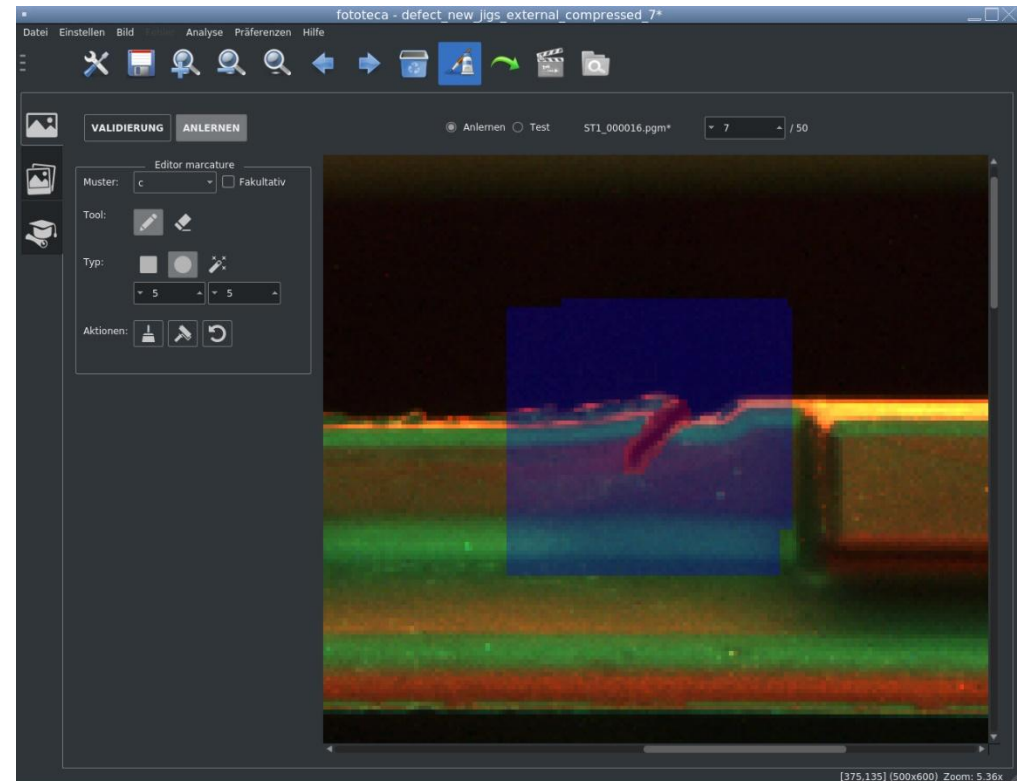
Demo

- Take pictures of the defects



Demo

- **Mark the defect** with the marking tool in the software
- This is the labelling process in supervised learning



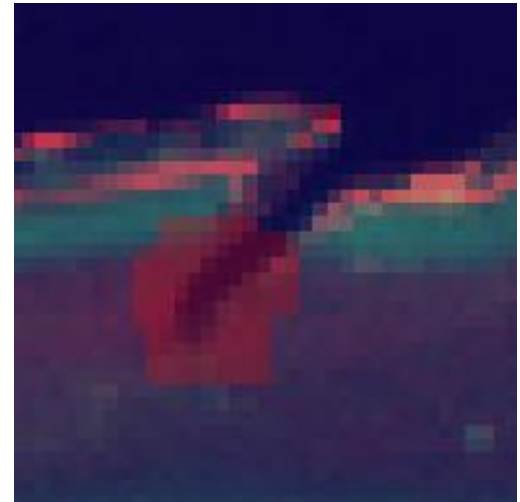
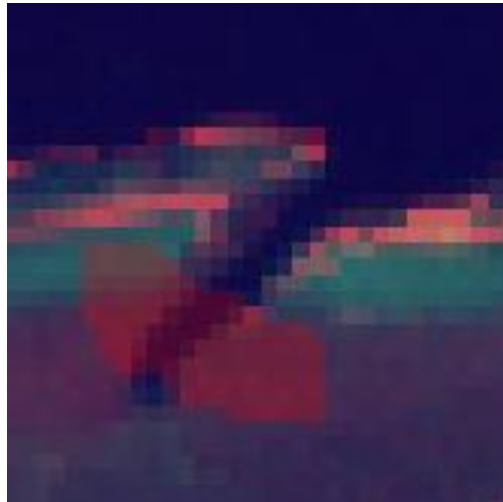
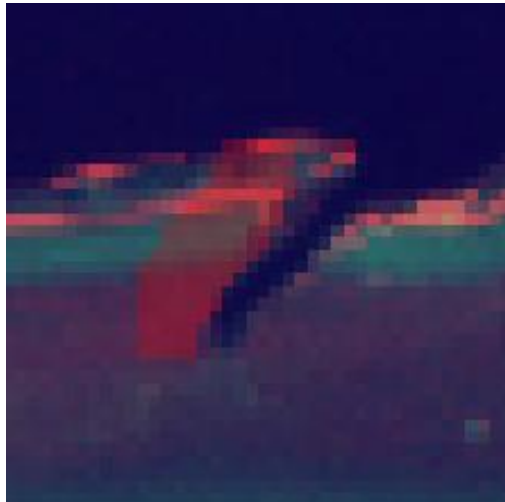
Demo

- **Iterate the marking process** until we have enough training samples
- Some examples of cutting defects



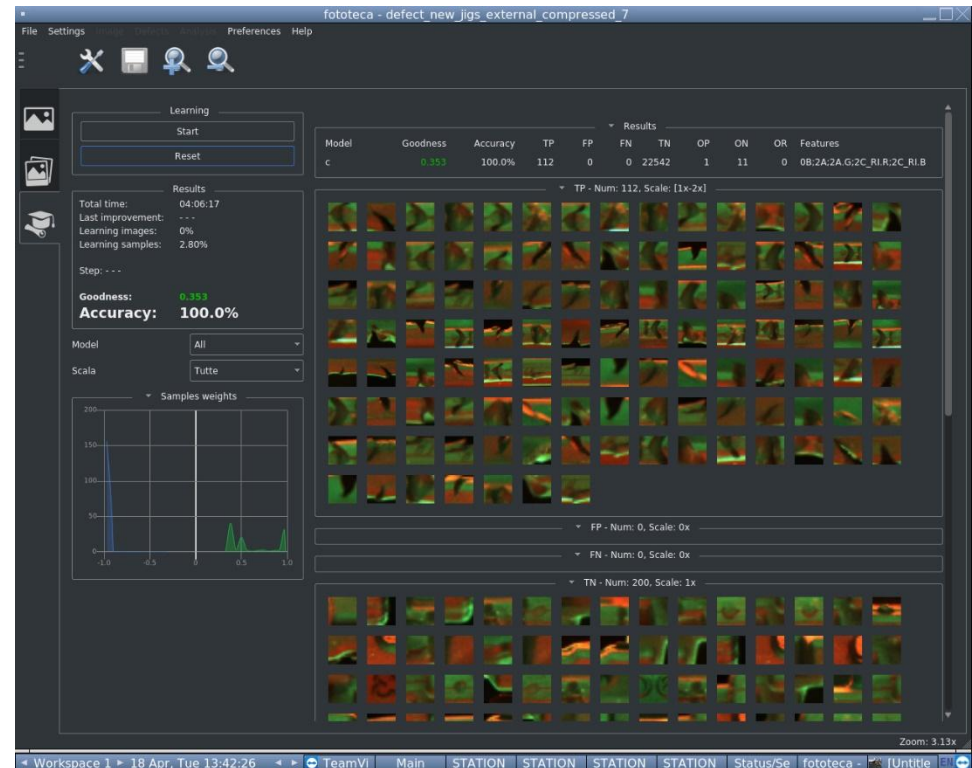
Demo

- **Avoid bad/missing marking**
- Avoid „Garbage in Garbage out“



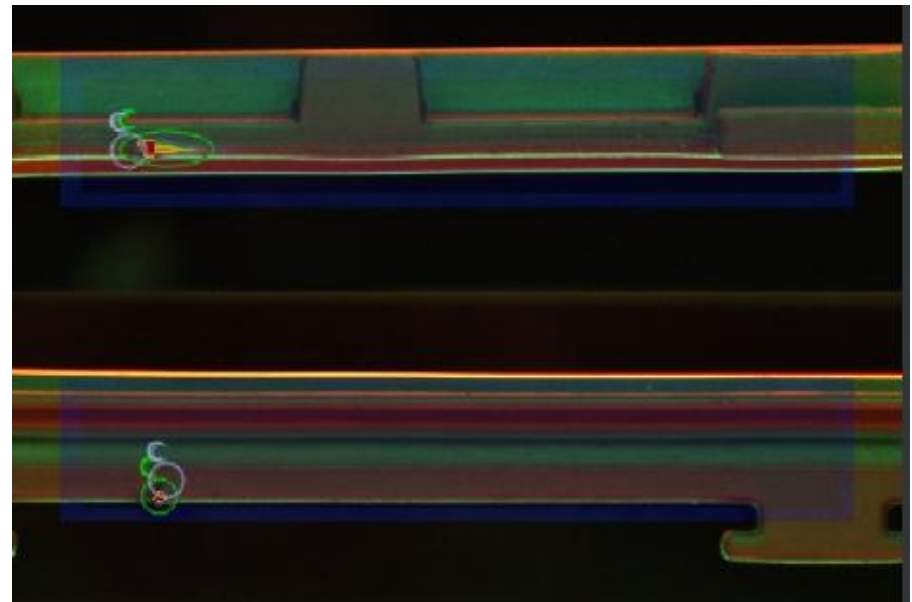
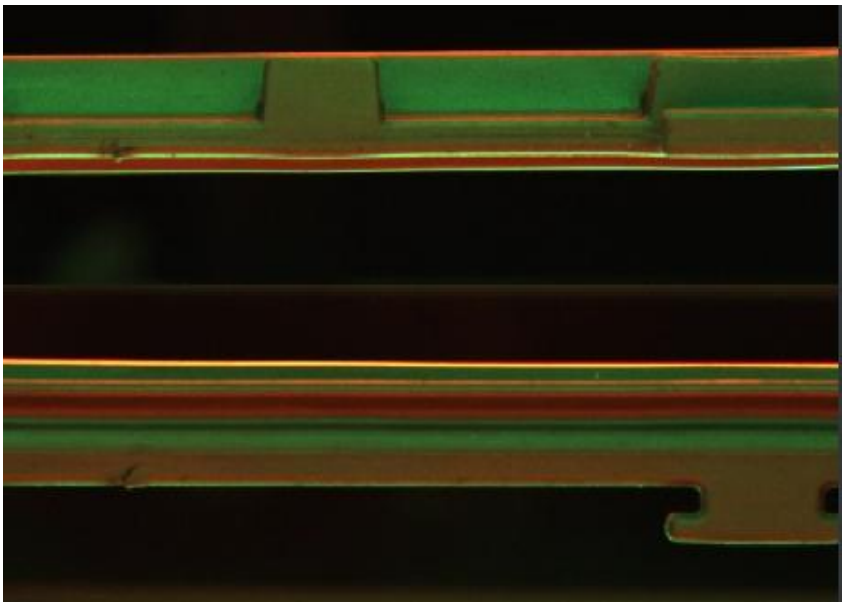
Demo

- Training interface
- Performance stats
- Adjust parameters
- Train until a satisfied result



Demo

■ Analysis after a successful training



Thank you!

Do you have any questions?